Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/01/2004
PI Name:	De Luca, Carlo Ph.D.		
Project Title:	Micro-Gravity Induced Changes in the Control of Muscles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cjd@bu.edu	**Fax:**	FY 617-353-5737
PI Organization Type:	UNIVERSITY	**Phone:**	617-353-9756
Organization Name:	Boston University - NeuroMuscular Research Center		
PI Address 1:	NeuroMuscular Research Center		
PI Address 2:	19 Deerfield Street, 4th Floor		
PI Web Page:	http://nmrc.bu.edu		
City:	Boston	**State:**	MA
Zip Code:	02215	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	06/01/2001	**End Date:**	05/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
Contact Email:	suzanne.g.mccollum@nasa.gov		
Flight Program:	Pre/Post Flight		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roy, Serge (Boston University NeuroMuscular Research Center)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	First, it will determine the neural component to the microgravity-induced deterioration in muscle function and strength, at both the micro and macro levels. This will assist in the design of techniques for monitoring rehabilitative interventions that target the specific alterations in muscle control. These interventions may include exercises that need to be performed in space and those that will be carried out upon return to Earth. Second, it will provide the knowledge base necessary to identify a ground-based model for the effects of microgravity (for example, the process of aging). The availability of a validated model for microgravity will facilitate the development of countermeasures by enabling the testing and validation of proposed countermeasures on Earth. Third, by comparing the effects of microgravity on muscles that are used to varying degrees in space, we will evaluate if low-level activity is useful in counteracting some of the effects of microgravity. This knowledge will be important in the design of countermeasures.		

	Finally, it is expected that the tracking of the recovery process in this study will identify the distinct effects of the two main the causes for microgravity- related losses in force: atrophy-based and neurally-mediated losses, since muscle force production is expected to recover from these following different timecourses. The knowledge regarding the significance of the contribution of the two components of loss will guide the design of countermeasures since different interventions may be developed to counter the effects of atrophy versus deterioration in neural control.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study aims to answer the basic scientific question regarding the effects of microgravity on the micro and macro control of muscle. By micro control we mean the control of motor units within a muscle; by macro control we mean the control of simultaneously acting muscles. With the information that it will provide on the effects of microgravity on muscle control, this study has the potential of providing various suggestions for the design of countermeasures. As for Earth Based research, the understanding of the neurological modifications that occur in microgravity should provide an accelerated time-scale model for the neurological effect on motor control due to aging. Additionally, the equipment that will be developed for studying the behavior of motor units can be used by neurologists in the clinical environment to assess the severity of diseases such as stroke, Parkinson"s, etc., as well as the success of various interventions.
Task Progress:	Our work on this project has made gains in the following areas: Decomposition Algorithms: The current decomposition algorithms are being adapted into a single software package that can be run on a laptop computer. This Decomposition Software Package is being integrated within the framework of the existing data acquisition hardware and software in order to complete a "stand alone" portable Decomposition System. In addition, a Processed Data Output Display Module is being developed to be used in conjunction with the Decomposition Software Package in order to display results in a graphical format. The above modules will be integrated within the shell of the existing User Interface so that they can be accessed and controlled by the operator's computer screen. Data Acquisition Hardware: We have been designing, and have completed a prototype for a new hand restraint device that will replace the previous device that did not comply with the test requirements for this project. Secondly, we are developing an improved fine-wire decomposition electrode. The existing electrode design has limitations both in its manufacture and performance. Currently, each fine wire electrode is fabricated by hand with the aid of a microscope. The technique is time consuming and requires training to achieve the high level of consistency needed for repeatable results. An improved fine wire electrode design is underway that should increase the chances for obtaining signals of suitable quality for decomposition analysis, and minimize the need for re-insertions. An electrode assembly fixture is being designed which will provide a means to accurately cut, position, and bond each of the individual electrode wires into a predefined array of contacts with spacing comparable to our "gold standard" needle electrode. We have also been developing surface electrode arrays that may replace the need for the current reliance on invasive needle and fine-wire recordings for this project. A prototype array was implemented and is being used to acquire data that can be analyzed for EMG decomposition. Data Acquisition Protocols: We have been conducting "dry-runs" in our laboratory in anticipation of the test procedures intended for crew testing.
Bibliography Type:	Description: (Last Updated: 12/18/2006)
Abstracts for Journals and Proceedings	Adam, A.; De Luca CJ "Motor unit control in human vastus lateralis muscle during fatigue." 30th Annual Northeast Bioengineering Conference Springfield, MA , Apr-2004
Abstracts for Journals and Proceedings	Nawab, S. H.; Wotiz, R.; De Luca, C. J. "Resolving EMG Pulse Superpositions Via Utility Maximization" 8th World Multiconference on Systemics, Cybernetics and Informatics Orlando, Florida , Jul-2004
Abstracts for Journals and Proceedings	De Luca, C.J.; Nawab, S.H.; Adam, A.; Roark, R.M.; Manto, M.U., Wotiz , R.; Srivastava, R.; Gilmore, L.D. "The Second-Generation Precision Decomposition System For EMG Signals: An Overview" XVth ISEK Congress Boston, MA , Jun-2004
Abstracts for Journals and Proceedings	Adam, A; De Luca, C.J. "Motor Unit Control In Vastus Lateralis Muscle During Fatigue." XVth Congress of ISEK Boston, MA , Jun-2004
Abstracts for Journals and Proceedings	Manto, M.U.; Sauvage, C.; Adam, A.; Roark, R.M.; Nawab, S.H.; De Luca, C. "Motor Unit Firing Patterns In Patients With Cerebellar Ataxia" XVth Congress of ISEK Boston, MA , Jun-2004
Abstracts for Journals and Proceedings	Roark, R.M.; Adam, A.; Zalvan, C.H.; Manto, M.U.; Nawab, S.H.; De Luca, C.J. "Clinical Evaluation Of Laryngeal Muscle Dysfunction Using Advanced EMG Decomposition" XVth Congress of ISEK Boston, MA , Jun-2004